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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

BT2104 -1 – REACTION ENGINEERING

(For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.

Linear graph papers may be provided.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

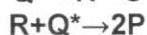
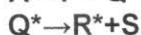
1. Rate of a chemical reaction is independent of the concentration of the reaction for a _____ reaction
A) Third order
B) First order
C) Zero order
D) Second order
2. Rate determining step in a reaction consisting of a number of steps in series is the _____ step
A) Slowest
B) Fastest
C) Intermediate
D) Data insufficient; can't predicted
3. Sum of the powers of the concentration terms in the rate equation is called the _____ of the reaction
A) Molecularity
B) Rate
C) Order
D) Overall Order
4. If 'n' is the order of reaction, then unit of rate constant is
A) $(\text{time})^1(\text{Concentration})^{1-n}$
B) $(\text{time})^{-1}(\text{Concentration})^{1-n}$
C) $(\text{time})^{-1}(\text{Concentration})^{n-1}$
D) $(\text{time})^{-1}(\text{Concentration})^{1+n}$
5. Free Energy change at equilibrium is
A) Zero
B) Positive
C) Negative
D) Indeterminate
6. Number of components (C), phase (P) and degree of freedom (F) are related by Gibbs phase rule as
A) $P + F - C = 2$
B) $C = P - F + 2$
C) $F = C - P - 2$
D) $P = F - C - 2$
7. Enthalpy 'H' is defined as
A) $\Delta H = \Delta U - \Delta PV$
B) $\Delta H = \Delta U + P\Delta V$
C) $\Delta H = \Delta U - P\Delta V$
D) $\Delta H = \Delta U + \Delta PV$
8. The standard free energy change and equilibrium constant are related as
A) $\Delta G = RT \ln K$
B) $\Delta G = -RT \ln K$
C) $\Delta G = -RT \ln K$
D) $\Delta G = RT \ln K$
9. Rate constant 'k' and absolute temperature 'T' are related by collision theory as
A) $k \propto T$
B) $k \propto e^{-(E/RT)}$
C) $k \propto T^{1.5}$
D) $k \propto \sqrt{T}$

10. A second order reaction of the form $A+B \rightarrow R$ is called pseudo-first order reaction, when
 A) $C_{A0} = C_{B0}$ B) $C_{A0} > C_{B0}$
 C) $C_{A0} < C_{B0}$ D) $C_{A0} > C_{B0}$
11. The reaction with low activation energies is
 A) Slow B) Always spontaneous
 C) Fast D) Always non-spontaneous
12. For every 10°C rise in temperature, the rate of chemical reaction doubles. When the temperature increases from 30 to 70°C , the rate of reaction increases _____ times.
 A) 8 B) 12
 C) 16 D) 32
13. Space time in flow reactor is
 A) The reciprocal of the space velocity B) Usually equal to the residence time
 C) A measure of its capacity D) Both (A) and (B)
14. For a first order reaction carried out in PFR, the space time is
 A) $\frac{1}{k} n \frac{C_o}{C}$ B) $k n \frac{C_o}{C}$
 C) $\frac{1}{k} n \frac{C}{C_o}$ D) $k n \frac{C}{C_o}$
15. For a mixed flow reactor operating at steady state, the rate of reaction is given by
 A) $\frac{F_{A0}}{V} - \frac{dC_A}{dt}$ B) $\frac{F_{A0}}{V} + \frac{dC_A}{dt}$
 C) $-\frac{dC_A}{dt}$ D) $\frac{F_{A0}}{V} \cdot X_A$
16. 'n' number of PFR in series with a total volume 'V' gives the same conversion as one PFR of volume
 A) V/n B) V
 C) $V \cdot n$ D) $1/V$
17. The 'E' curve for a non-ideal reactor defines the fraction of fluid having age between t and $t + dt$ _____?
 A) At the inlet B) At the outlet
 C) In the reactor D) Averaged over the inlet and outlet
18. The residence time distribution of an ideal CSTR is
 A) $\frac{1}{\tau} e^{(-t/\tau)}$ B) $\tau \cdot e^{(-t/\tau)}$
 C) $e^{(-t/\tau)}$ D) $\frac{1}{\tau} e^{(t/\tau)}$
19. Exit age distribution and $F(t)$ are related as _____
 A) $dF/dt = E$ B) $E \times V = F$
 C) $dE/dt = F$ D) $F \times t = E$
20. Stimulus-response techniques are commonly used to characterize the extent of non-ideal flow in vessels. Tracer input signal is used as stimulus. Any material can be used
 A) as tracer if it can disturb the flow pattern in the vessel B) as tracer if it does not disturb the flow pattern in the vessel and it can be detected
 C) as tracer D) as tracer if it follows ideal flow patterns

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) Define reaction. With suitable examples explain the classification of a chemical reaction.	8	L*2	1	1
	b) How is the equilibrium constant K related to the standard free energy change? Does K vary with pressure?	8	L2	2	2

2. a) Prove that the below mechanism can explain the first order decomposition of substrate A.



Where A is a substrate, P & S are products, R* and Q* are intermediates.

- b) What is Gibb's free energy? How is spontaneity of a reaction determined by Gibb's free energy? Explain.

Unit - II

3. a) Explain the significance of activation energy.
b) The decomposition of Nitrous pent oxide in Carbon tetrachloride solution is carried out at 48°C in a batch reactor yields following data, show that the decomposition is first order.

Time, min	10	15	20	25	∞
Volume of O ₂ evolved, mL	6.3	8.95	11.4	13.5	34.75

4. a) Describe an initial rate method of interpreting kinetic data.
b) Derive an integral expression in terms of concentration for bimolecular irreversible reaction of the type $A+B \rightarrow R+S$.
5. a) Differentiate between molecularity and order of reaction.
b) Derive an expression to determine the overall order of an irreversible reaction from half-life data. Also calculate the order for the reaction $2A+2B \rightarrow R+2S$ from the following constant volume data obtained by using equimolar quantities of A and B at 296K.

Half Life, sec	265	186	115	104	67
Total Pressure, mmHg	200	240	280	320	360

Unit - III

6. a) Derive an expression to determine the batch time required to degrade the biohazard compound from initial concentration C_0 to final concentration C_f .
b) What is C, E and F curve? Explain the significance of C, E and F curve in determining the degree of non-ideality in reactor.
7. a) The degradation of dye from the paint shop of an automotive industry is carried out in a MFR. The holding time and extent of degradation are given below.

Holding Time, Sec	0.423	5.1	13.5	44	192
Extent of degradation	0.22	0.63	0.75	0.88	0.96

If the initial concentration of dye in the effluent is 0.002mol/L, determine the kinetics of degradation.

- b) Obtain an expression for finding the mean residence time in mixed flow reactor.
8. a) The exothermic reaction was carried out adiabatically and the following data recorded. The entering molar flow rate of reactant was 10mol/s.

XA	0	0.2	0.4	0.6	0.8
-r _A , mol/L. s	0.182	0.143	0.10	0.0667	0.0357

What are the PFR and CSTR volume necessary to achieve 60% conversion?

- b) The data given below represent a continuous response to a pulse input into a MFR which is to be used as for the degradation of dye. Calculate the mean residence time of fluid in the vessel, $\bar{t}$ and tabulate and construct the E curve.

Time, min	0	2	4	6	8	10	12	14
C, g/m ³	0	5	10	6	3	1.5	0.6	0

8 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
